Fenil Miteshkumar Modi

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Research Interests

Orbital data analysis and space situational awareness; probabilistic conjunction assessment, covariance fusion and uncertainty quantification; large-scale scientific data processing and reproducible analysis pipelines; statistical evaluation of randomness and cryptographic primitives; scientific data visualisation.

Education

Silver Oak University — Ahmedabad, India
Bachelor of Technology, Computer Engineering — *Aug 2022 – Aug 2026*

- **Status:** all coursework and examinations completed August 2026; degree conferral pending.
- **Relevant coursework:** Statistical Methods, Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Cryptography, Computer Networks, Database Management Systems, Scientific Computing.
- **Self-directed study:** orbital mechanics (two-body dynamics, Keplerian elements, SGP4/SDP4 propagation), numerical methods, probabilistic modelling and Bayesian inference (in progress).
- **Final-year work:** probabilistic conjunction assessment under positional uncertainty — basis of the IEEE-published ICSSIT 2026 paper.

Publications

Modi, F. M., Khara, S. V., Tivari, G. D., Patel, J., Patel, P., & Kumawat, G. (2026). *Probabilistic Space Debris Conjunction Assessment using Machine Learning and Covariance Intersection Analysis.* In *Proceedings of the International Conference on Sustainable Systems and Innovative Technologies (ICSSIT 2026).* IEEE. Technically co-sponsored by the IEEE Systems, Man and Cybernetics Society. DOI: [[DOI]]. ieeexplore.ieee.org/document/11656410

Contribution (first author): led the study design and analysis — fusing state estimates of unknown cross-correlation via covariance intersection and applying machine learning to conjunction screening, showing through B-plane projection of 3σ covariance ellipses that a nominal miss distance can remain statistically consistent with collision. Validated against the Iridium 33 / Cosmos 2251 event.

Conference Presentations

Probabilistic Space Debris Conjunction Assessment using Machine Learning and Covariance Intersection Analysis. Oral presentation delivered by **F. M. Modi** on behalf of all authors, ICSSIT 2026, 28–30 July 2026 (online).

Research Experience

ORCAS — Orbital Risk and Conjunction Assessment System — Ahmedabad, India
Team Lead & Lead Analyst, Silver Oak University — *Jan 2024 – Present*

- Cut catalogue preparation from a manual multi-hour task to a single automated pass, as measured by end-to-end runtime over a ∼10,000-object TLE catalogue, by building a Python ingestion pipeline validating checksums, resolving epoch drift and de-duplicating records.
- Characterised the tracked-object population by orbital regime, as measured by element distributions across LEO/MEO/GEO and inclination bands, by performing exploratory data analysis on the cleaned catalogue.

- Narrowed the conjunction search space ahead of expensive computation, as measured by candidate object pairs surviving each filter stage, by implementing a coarse-to-fine screening cascade with perigee–apogee and geometric pre-filters.
- Replaced point estimates with quantified confidence, as measured by 95% confidence intervals on probability of collision, by running Monte Carlo sampling over positional covariance as a batch-parallel workload.
- Established propagator correctness before publication, as measured by position residuals against published reference ephemerides over a multi-day horizon, by cross-validating the SGP4 implementation against independent reference outputs.
- Coordinated a six-member research team — the co-author group of the ICSSIT 2026 paper — keeping work reproducible, as measured by full reconstruction of every published figure from versioned source, through Git-based code review and parameterised analysis notebooks.
- *Stack:* Python, SGP4/Skyfield, NumPy, SciPy, Pandas, Scikit-learn, FastAPI, React.

Quantum-Enhanced AES Encryption using Quantum Random Walks Ahmedabad, India
Independent Research Project, Silver Oak University *Aug 2023 – Dec 2023*

- Determined whether quantum-derived keys outperform pseudo-random streams, as measured by pass rates across the 15 statistical tests of NIST SP 800-22, by designing a controlled comparison over matched-length key sets.
- Characterised diffusion behaviour with stated significance levels, as measured by χ^2 goodness-of-fit and avalanche-effect distributions, by implementing entropy and bit-correlation analysis in NumPy/SciPy.
- Made the study independently repeatable, as measured by full regeneration of results from a documented seed and parameter set, by recording experimental methodology and data-collection procedure alongside the code.
- *Stack:* Python, Qiskit, NumPy, SciPy, Matplotlib.

Professional Experience

Hubbl — Technology Community Platform India
Lead Backend & Data Engineer (Full-time) *Jul 2026 – Present*

- Defined the backend and data architecture at platform inception — service boundaries, data contracts and the relational schema underpinning all community, event and speaker records — as measured by 8+ core backend services, for a platform consolidating four regional technology user groups (Gujarat Tableau, Gujarat Databricks, Ahmedabad Tableau, India AI + Tableau) into one ecosystem.
- Turned fragmented per-group records into one queryable dataset, as measured by four previously separate user-group datasets normalised into a single PostgreSQL schema, by designing the data model and building the ingestion path with Prisma and Zod-validated contracts.
- Kept community and event data current without manual entry, as measured by mail-synchronisation and event-scraping jobs running unattended on schedule, by building Node-based background workers against IMAP inboxes and external community sources.
- Made privileged actions traceable across the platform, as measured by an audit log covering role-stage and role-review state transitions, by building role management and profile change-request handling into the service layer.
- Delivered the registration data pipeline for a 120-capacity national analytics event run with the India AI + Tableau User Group and Queen's University Belfast (India GIFT City) on 17 August 2026, as measured by a fully pre-registered cohort requiring no on-the-spot fallback, by owning participant records and registration analytics end to end.
- *Stack:* Node.js, Express.js, PostgreSQL (Supabase), Prisma, Zod, JWT + bcrypt, REST APIs, SMTP/IMAP, scheduled background jobs; deployed on Render behind Cloudflare.

Additional Projects

Diablex — Clinical Time-Series Analytics Platform Independent
Full-Stack Developer *2024 – Present*

- Reduced noise-driven false anomaly flags in continuous glucose data, as measured by flag counts before and after filtering across ∼1,000 readings per day, by implementing outlier detection, resampling and normalisation for irregular time series.
- Enabled clinicians to distinguish signal from instrumentation artefact, as measured by data-quality flags attached to every derived statistic, by surfacing provenance and confidence alongside each reported trend.
- *Stack:* Node.js, React, MongoDB, Python.

Retail Sales Performance Dashboard Independent
Data Analysis Project *2026*

- Delivered an executive dashboard covering four KPI tiles and three linked views, as measured by drill-down coverage across region, category and customer segment, by modelling table relationships and wiring dashboard actions for cross-filtering.
- Surfaced under- and over-performing segments at a glance, as measured by variance against target in a diverging bar view, by designing the encoding around deviation rather than absolute value.
- *Stack:* Tableau, data modelling, dashboard actions.

Technical Skills

Scientific Computing & Data Analysis: Python (NumPy, SciPy, Pandas), exploratory data analysis, time-series analysis, statistical hypothesis testing, Monte Carlo methods, uncertainty quantification, numerical integration

Large-Scale & High-Performance Computing: batch data-processing pipelines, catalogue-scale dataset handling, scheduled background jobs, parallel workloads, performance profiling, Linux/CLI workflows

Orbital & Space Data: SGP4/SDP4, TLE ingestion and validation, Skyfield, ephemeris cross-validation, conjunction screening, covariance intersection, B-plane geometry, coordinate frame transformations

Machine Learning: Scikit-learn, TensorFlow, PyTorch; cross-validation, residual analysis, hyperparameter tuning, model selection and evaluation

Programming: Python, C/C++, Java, JavaScript, SQL

Backend & Data Engineering: REST API design, service architecture, database schema design, indexing and query optimisation, JWT/bcrypt authentication, scheduled workers, third-party integration; Node.js, Express.js, PostgreSQL, Prisma, Zod, FastAPI, MongoDB, React

Visualisation: Matplotlib, Seaborn, PyVista, Tableau, D3.js

Research Practice: Git/GitHub version control, reproducible analysis, LaTeX, Jupyter, technical documentation, collaborative development in multi-member teams

Professional Development

India AI + Tableau User Group — Data & Analytics Day (with Queen's University Belfast, India GIFT City, and Hubbl). Attendee, 17 August 2026. Hands-on analytics sessions and industry-mentor networking; 120-attendee capacity, pre-registration only.

Presented project work in departmental reviews and college innovation challenges.

Leadership & Service

Team Lead, ORCAS research team, Silver Oak University (Jan 2024 – Present) — task allocation, code review and reproducibility standards across a six-member team.

Organised and participated in college-level technical events and hackathons.

Languages

English (full professional proficiency) · Hindi (native) · Gujarati (native)

Availability

Open to research positions, graduate traineeships and postgraduate study internationally. Available for relocation.

References

Available on request.